Data Structures
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Course & Reference Books

- Data Structures Using C++ (By D.S. Malik)
- Data Structures Using C and C++ (By Y. Langsam, M. J. Augenstein, A. M. Tenenbaum)
- Data Structures and Algorithms (By A. V. Aho, J. E. Hopcroft, J. D. Ullman)
- Schaum's Outline Series, Theory and problems of Data Structures (By Seymour Lipschutz)

Office Hours

- Mon to Thursday (11:30 am to 2:00 pm)
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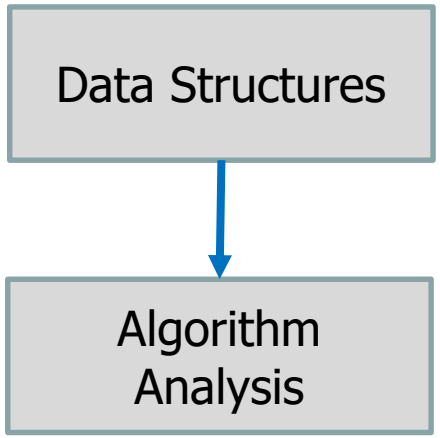
Tentative Evaluation Breakdown

Evaluation Name	Weightage
Quiz	10
Assignment	15
Project	10
Sessionals	25
Final	40

Course Content

- Introduction to data structures and abstract data types
- Arrays
- Background: Templates and Algorithm Analysis
- Searching and sorting techniques
- Linked Lists
- Queues
- Stacks
- Trees: Definitions and terminology
- Trees: Binary Search Trees, AVL Trees
- Heap
- Graphs
- B+ Trees
- Hashing

General Overview



How to organize information/data, usually in memory, so that it can be used efficiently?

How to analyze the performance of algorithms to evaluate their suitability for solving complex problems?

What is a Data Structure?

- In a general sense, any representation that is used for storing information is a data structure
  - Example: integers, structures, classes, arrays, etc.

- More typically, a data structure provides a way for organizing a collection of data items
  - Storing, accessing and modifying data items

- Most data structures have associated algorithms to perform operations on data
  - Search, insert, remove etc.

Costs and Benefits

- Each data structure requires
  - Space for storing data items
  - Time to perform each basic operation
  - Programming efforts

- Cost of a data structure
  - Time and space resources it consumes

- Choice of data structure depends on many factors
  - Type of data
  - Frequency with which various data operations are applied

- Hard to define data structure that performs better in all situations
  - Time and space tradeoff

Goals of This Course

- Learn the commonly used data structures
  - Form a programmer's basic data structure "toolkit"
- Case studies of data structures
- Examine the costs and benefits of every data structure

Motivational Example

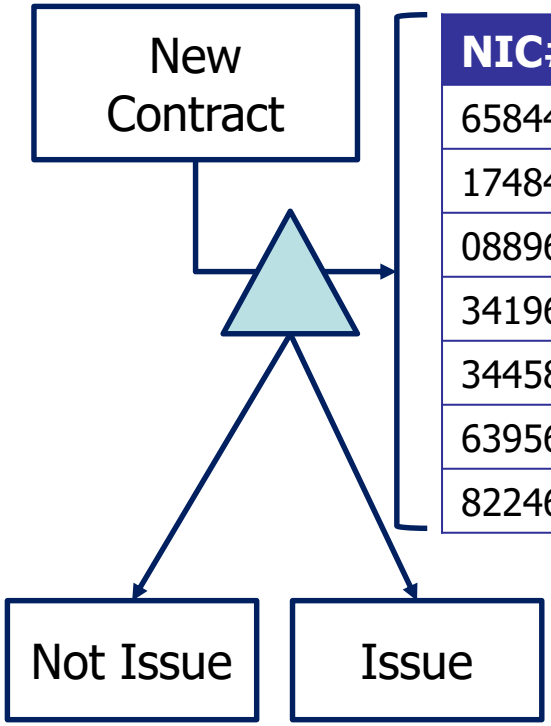
- A cellular service company provides contract to its 10 million users
- Due to new security enforcements, the company wants to prevent issuing of multiple contracts to users
- Method of Detecting Multiple Contracts
  - Before issuing a new contract to user
  - First search the id of user in existing contracts database
  - In case of failure → issue a new contract
  - In case of success → do not issue a new contract to user

Example: Linear Array Data Structure (1)

NIC#	Name	Address
6584495-9	Muhammad Faheem	House No 3 Gulshan Bahar Sec 16
1748425-5	Naeem Alam	A-11 Shams Plaza Block-B N.Nazimabad
0889679-1	Arslan Akhtar	H No 152 Bostang Colony
3419668-1	Zain Ahmed	Sharfabad Street Gulshan Karachi
3445864-3	Sumair Farooq	Post Office Tayyar, Multan
6395653-4	Ali Affan	H.No. 425, Sector F-11/4, Islamabad
8224641-1	Syed Faraz	Sharfabad Street Gulshan, Faisalabad

- Linear Array (with 10 million entries)
  - 3 arrays (NIC, Name, Address)
  - Structured array
  - Class's object array

Example: Linear Array Data Structure (2)



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- Any disadvantage of linear array data structure?
- How to improve?

Example: Improved Data Structure

- Create a dictionary data structure
  - Group similar records together
  - Similarity in terms of first digit of NIC number
  - Add a dictionary entry for each distinct digit (0 – 9)

- Example: 3419668-1, 3445864-3, 1748425-5.
  - 3 and 1 are dictionary entries

- Existing contracts are searched in two steps
  1. Search the dictionary entry (i.e., group searching)
  2. Search contract within the group (i.e., with the same NIC digit)

- How much improvement w.r.t. linear array?

Example: Dictionary Data Structure (1)

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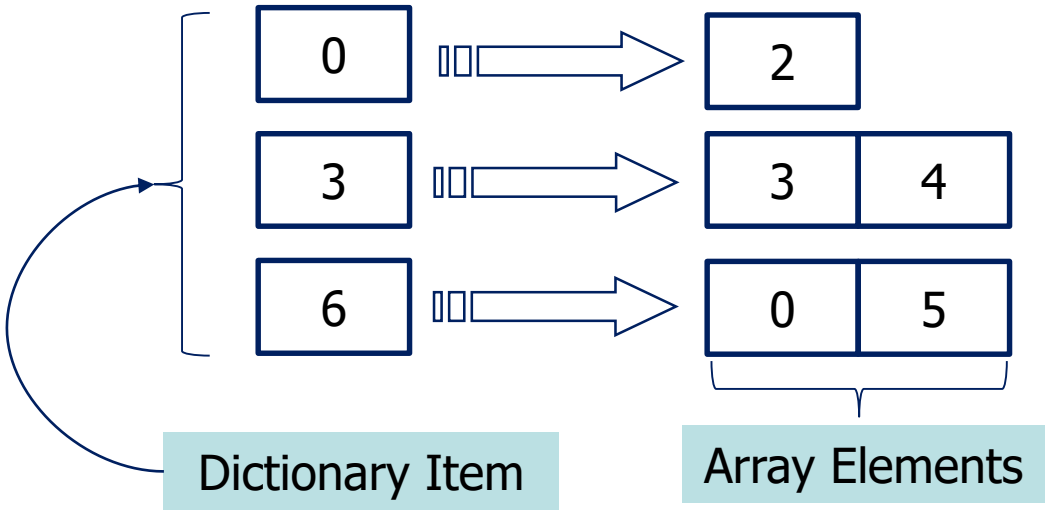
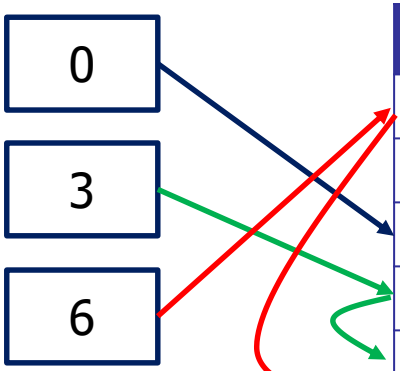


Example: Dictionary Data Structure (2)

- Another possibility
  - Maintain pointers with structures (or records)
  - Non NULL pointer indicates presence of next record



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Practice Task

- Implement the example on the previous slides in C++ in two different programs. In the first, implement the data structure as a single array. In the second, implement the dictionary approach where the data is indexed by the first digit of the CNIC. In both programs, generate a random CNIC and search for it in your data structure. Do this searching operation in a loop of 500, to simulate 500 searches.
- Now, one by one, run both programs and note the running time. Use a Linux utility for checking the running time. Submit your programs and your result of running time for both (in a separate txt file).

What is Data Structure Efficiency?

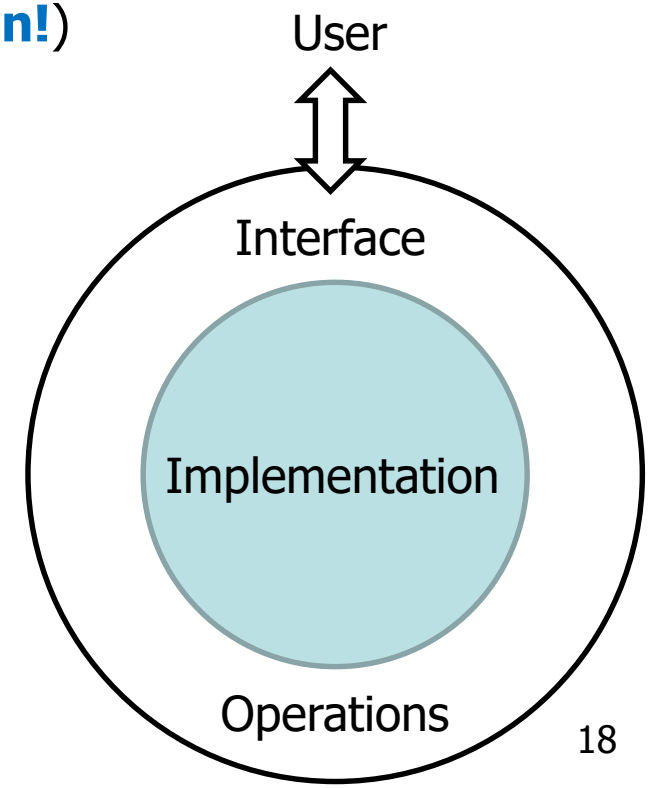
- Two major measures of efficiency of a data structure or algorithm
  - Time and space
- Cost of data structure
  - Time and space resources it consumes
- Efficiency of a data structure depends on its operations
  - Makes a difference between program running in few seconds or days
- Difficult to define data structure that performs better in all situations

Abstract Data Types (1)

- A definition of data type solely in terms of
  - Set of related data items (or values)
  - Set of operations on the data

- Separation of logical properties from the implementation details
  - Hide implementation details (**Encapsulation!**)

- What not how is focus



Data Structures vs. ADT

- A data structure is physical implementation of an ADT
  - Each operation associated with ADT is implemented by one or more subroutines in the implementation

- Data structure usually refers to an organization of data in main memory

- Example: Stack, or Queue are ADTs, but a C++ Queue implemented in a particular class with particular functions is a concrete data structure.

- The relationship between ADT and Data Structure is a bit like the relationship between an interface (class.h file – tells WHAT is done) and a class implementation (the class.cpp file – tells HOW it is done).

Array as an ADT

- An array is defined as
  - Ordered collection of a fixed number of elements
  - All elements are of the same data type

- Basic operations
  - Direct access to each element in the array
  - Values can be retrieved or stored in each element

C/C++ Implementation of an Array ADT

```
datatype arrayName[intExp];
```

As an ADT	In C/C++
Ordered	Index: `0,1,2, … SIZE-1`
Fixed Size	`intExp` is constant
Homogeneous	`dataType` is the type of all elements
Direct Access	Array subscripting operator `[ ]`

Summary

What is a data structure?

- In a general sense, any representation that is used for storing information is a data structure
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Where is a data structure helpful?

- Data structure can make the difference between a program running in a few seconds or many days
  - Efficiency of data structure

- **What is an ADT?**
  - A logical representation of a data structure.

Any Question So Far?